

Bell Local vs Quantum Simulation

Trials per setting: 100000

Local Hidden Spin Model

This model uses shared hidden directions and local measurements.
It does NOT use the quantum probability rule.

Settings A, B:
same results = 24986
opposite results = 75014
measured E = -0.500

Settings A, B':
same results = 25057
opposite results = 74943
measured E = -0.499

Settings A', B:
same results = 24955
opposite results = 75045
measured E = -0.501

Settings A', B':
same results = 75065
opposite results = 24935
measured E = 0.501

S = -2.001

|S| = 2.001 - A finite simulation may introduce random sampling noise while increasing the number of trials will slow down the script.

Quantum Singlet Probability Model

This model uses the quantum spin-singlet probability rule:
 $E = -\cos(\theta)$.

Settings A, B:
same results = 14678
opposite results = 85322
measured E = -0.706

Settings A, B':
same results = 14719
opposite results = 85281
measured E = -0.706

Settings A', B:
same results = 14898
opposite results = 85102
measured E = -0.702

Settings A', B':
same results = 85590
opposite results = 14410
measured E = 0.712

S = -2.826

|S| = 2.826 - Within the interpretation proposed in this work, this reflects the angular dependence of the quantum correlation function $E(\theta) = -\cos\theta$.